

Interstellar Route Construction Plan

Background

In the era of interstellar colonization, you are an engineer responsible for building the interstellar transportation network. There are n planets in this galaxy (numbered from 0 to $n-1$) and m potential interstellar routes between them. Each route i bidirectionally connects planets u_i and v_i , and has an initial **travel capacity** s_i .

You need to select a set of routes to build and possibly upgrade so that all planets become connected, forming a stable transportation network.

Special Routes: Wormholes

Some routes contain natural wormholes (marked as $must_i=1$). These wormhole routes are extremely valuable and **must be included** in the final network. Due to their special structure, **they cannot be upgraded**.

The remaining routes ($must_i=0$) are ordinary routes, and you are allowed to **upgrade** them. Each upgrade operation **doubles** the travel capacity of that route. A single route can be upgraded multiple times, but the total number of upgrades across all routes cannot exceed your resource limit k .

Network Stability

The **stability** of the entire transportation network is defined as the minimum travel capacity among all selected routes. Your goal is to maximize this stability.

Task

Compute the ****maximum possible stability**** under the following conditions:

1. The selected routes must connect all n planets.
2. All wormhole routes (those with $must_i=1$) **must be included** and **cannot be upgraded**.
3. You may perform at most k upgrade operations in total. Each operation selects an ordinary route and doubles its travel capacity ($s \leftarrow s \times 2$). A route can be upgraded multiple times.

If it is impossible to connect all planets under any circumstances, output -1 .

Input Format

- The first line contains three integers n, m, k , where n is the number of planets, m is the number of available routes, and k is the maximum number of upgrades allowed.
- The next m lines each contain four integers $u_i, v_i, s_i, \text{must}_i$, describing a route. Here, u_i and v_i are the endpoints, s_i is the initial travel capacity, and must_i indicates whether the route contains a wormhole.

It is guaranteed that there is at least one route with $\text{must}_i=1$.

Output Format

Output a single integer T , the maximum achievable stability.

If it is impossible to connect all n planets even using all available routes, output -1 .

Sample Data

Sample Input 1

```
4 5 2
0 1 6 1
0 2 1 0
0 3 2 0
1 2 5 0
1 3 1 0
```

Sample Output 1

```
5
```

Sample Input 2

```
4 5 5
0 1 9 1
0 3 2 0
1 2 5 0
1 3 1 0
2 3 3 1
```

Sample Output 2

```
3
```

Constraints (for reference)

- $2 \leq n \leq 10^5$
- $n-1 \leq m \leq 2 \times 10^5$
- $0 \leq k \leq 10^9$
- $1 \leq s_i \leq 10^9$
- $must_i \in \{0, 1\}$